PHYSICS QUESTION SERIES

Time: 15 Minutes **Maximum Marks:** 22
Question Type: Objective (MCQ) **Set:** Newton's Laws set 1

Instructions

- Each question carries equal marks.
- Only one option is correct.
- No calculators or electronic devices are allowed.

1. A person has a mass of 60 kg on Earth. What will be the weight of this person on the Moon? (Consider g_{Earth}=10 m/s^2 and $g_{moon} = \frac{g_{Earth}}{6}$)

   (A) 10 N
   (B) 100 N
   (C) 60 N
   (D) 600 N

2. If you drop a ball from the roof of a house and takes 10 min to reach the ground on the Earth. Neglecting air resistance, how much time will the same ball take to reach the ground when dropped from the same height on the Moon?

   (A) 4.1 min
   (B) 10 min
   (C) 24.5 min
   (D) 60 min

3. An exoplanet has twice the mass and twice the radius of the Earth. What will be the acceleration due to gravity on the surface of this exoplanet? (Take acceleration due to gravity on Earth as 10 m/s^2.)

   (A) 2.5 m/s^2
   (B) 10 m/s^2
   (C) 20 m/s^2
   (D) 5 m/s^2

4. If the Earth were to move to an orbit such that its distance from the Sun becomes twice its present value, what would be the approximate number of days in one year on Earth? (Assume the length of a day remains unchanged.)

(A) 365 days
(B) 1030 days
(C) 520 days
(D) 730 days

5. Since light cannot escape from a black hole, the escape velocity from its surface must be equal to the speed of light. If an Earth-sized object were to become a black hole, how many times greater than the Earth's mass would its mass need to be?(Consider escape velocity at the surface of the Earth as 10 m/s and the speed of light as $c = 3 \times 10^8$ m/s.)

(A) Infinity
(B) 3×10^{14}
(C) 9×10^{16}
(D) 9×10^{14}

6. After escaping the Earth's gravitational field, a rocket moves with a constant speed of 10 km/s directly towards the Moon. Approximately how much time will it take for the rocket to reach the surface of the Moon?

(Take the average distance between the Earth and the Moon as 3.84×10^8 m.)

(A) 6 hours
(B) 24 hours
(C) 10.5 hours
(D) 4.4 days

7. A mercury barometer is taken to the surface of the Moon. What will be the height of the mercury column in the barometer?

(A) 76 cm
(B) 12.7 cm
(C) 456 cm
(D) 0 cm

8. The Moon revolves around the Earth in a nearly circular orbit with orbital speed v. By what percentage should its orbital speed be increased so that the Moon can escape the Earth's gravitational field?

(A) 41%
(B) 6%
(C) 50%
(D) 100%

9. The Moon revolves around the Earth in a circular orbit with speed v. If, without changing its position or direction of motion, the Moon's speed is suddenly increased by 50%,what will be the subsequent nature of the Moon's motion? (Neglect the gravitational effect of other celestial bodies.)

   (A) It will move in a larger circular orbit around the Earth
   (B) It will move in an elliptical orbit around the Earth
   (C) It will escape from the Earth's gravitational field along a parabolic path
   (D) It will escape from the Earth's gravitational field along a hyperbolic path

10. In a Gaussian gun, the final steel ball is observed to emerge with a speed much greater than that of the initially fired ball. This observation is best explained by the fact that:

   (A) Newton's laws of motion are violated at high magnetic field strengths
   (B) Magnetic potential energy stored in the magnet–ball system is converted into kinetic energy of the balls
   (C) Momentum conservation does not apply to systems involving magnetic forces
   (D) The magnet continuously generates energy through electromagnetic induction

11. A rattleback placed on a rough horizontal surface reverses its direction of rotation when spun in one direction, but continues to rotate smoothly when spun in the opposite direction.

   Which of the following statements must be true for this directional reversal to occur?

   (A) Friction between the rattleback and the surface must be zero
   (B) The center of mass of the rattleback must lie exactly at its geometric center
   (C) There must exist a coupling between rotational motion and oscillatory motion due to asymmetry
   (D) The net external torque on the rattleback must always be zero

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ANSWER KEY

Physics MCQ Test
Newton's Laws set 1

Q. No.	Correct Option
1	B
2	C
3	D
4	B
5	D
6	C
7	D
8	A
9	D
10	B
11	C

Need Help with the Answers?

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